

Wine Production & Execution Sequence

RDD (Douro 1933–2022) · RVV (Vinho Verde 1941–2022) · Distribution Rules via CarenR



METEOROLOGICAL

BCM_base_*.xls
dados_finais_*.xlsx
Daily Tmax, Tmin, Tmed, Rain, ETo



PRODUCTION + PHENOLOGY

Pheno_Prdo_*.xlsx
Annual Wine_mhl + DOY of BB, FI, sM, Fs, Hv

STEP 1 · DATA PREPARATION

1 Data Cleaning & Binary Indicators

data_preparation_stage_1.R

Reads raw Excel files for both regions. Computes derived binary/threshold indicators and standardises column names. Outputs one CSV per region for meteorological data and production/phenology data.

INPUTS

BCM_base_*.xls

dados_finais_*.xlsx

Pheno_Prdo_*.xlsx

PROCESSING

Column standardisation

Tmed<15 / Tmax>35 / Tmin<0
flags

Sm<1.5Wp / Sm>0.9Fc flags

Date → Day / Month / Year

OUTPUTS

dataPrep_meteo_rdd.csv

dataPrep_meteo_rvv.csv

dataPrep_production_rdd.csv

dataPrep_production_rvv.csv

STEP 2 · FEATURE ENGINEERING

2 Phenological Window Features + Discretisation

features_creation_v2.R

For each year (y1) uses the DOY of each phenological stage as anchor. Computes climate aggregates in centred (c), before (b) and after (a) windows in both the current (y1) and prior (y0) year. All 59 features are then frequency-discretised (equal-frequency bins) for CarenR input.

INPUTS

dataPrep_meteo_*.csv

dataPrep_production_*.csv

WINDOW NOTATION

59 FEATURES BUILT (BY TYPE)

Mean Temp
tm_* (14 feats)

Days <15°C
tm.days.less.15 (2)

Frost days
days.minTemp.

Heat days
days.maxTemp.>35

DISCRETISATION

arules::discretize(method="frequency")

Constants (≤1 unique value) → factor

Commas → | (CarenR separator fix)

